Conway's Doomsday Algorithm

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The following document describes John Conway's 1973 ''Doomsday algorithm'' for determining the day of the week for any date.

1 Supplementary Definitions

We begin with supplementary definitions of propositions and their equivalence, the divisibility operator $x \mid y$, the flooring operator $\lfloor x/y \rfloor$,[1] and the modulo operator $x \,\%\, y$, for the reader's reference.

Definition 1.1 (Equivalence of propositions)**.** A proposition is a declarative statement that is either true or false, but not both. Propositions are equivalent if they share the same truth value. If two propositions x and y are *always* equivalent, we say ''x if and only if y'' and write $x \iff y$. Additionally, $(x \iff y) \iff (y \iff x)$.

Example 1.1 (Equivalence of propositions)**.** Let $x \in \mathbb{N}$ be a natural number.[2] The propositions ''x is even'' and ''x is not odd'' are equivalent because x must either be even and not odd, or it must be not even and odd: the statements always agree. It will never occur that x is both even and odd, or that it is neither even nor odd.

Definition 1.2 (Divisibility)**.** Let $a, b \in \mathbb{N}$ be natural numbers. We say that a is divisible by b, or equivalently, that b divides a, if there exists a natural number $k \in \mathbb{N}$ such that $a = k \cdot b$. If a is indeed divisible by b, we write $b \mid a$; otherwise, we write $b \nmid a$.

Example 1.2 (Divisibility)**.** The year $y = 1868$ is divisible by 4 because $1868 = 467 \cdot 4$, and 467 is a natural number. Note that divisibility by 4 or 100 depends only on the final two digits. For instance, in this example, 1868 is divisible by 4 because $68 = 17 \cdot 4$, and 17 is a natural number. A year is divisible by 100 if and only if its final two digits are both zeros. In this example, the year is clearly not divisible by 100. Another way to check divisibility by 4 is that $4 \mid y$ if and only if $y/2$ is even. For instance, in this example, the year 1868 is divisible by 4 because $68/2 = 34$ is even, noting that only the final two digits are relevant.

Definition 1.3 (Floor)**.** Let $a, b \in \mathbb{N}$ be natural numbers, called the *dividend* and *divisor*, respectively. The floor of a/b is defined as the unique natural number q satisfying

$$qb \leq a < (q+1)b, \tag{1.1}$$

and we write $q = \lfloor a/b \rfloor$. Intuitively, $\lfloor a/b \rfloor$ says ''divide a by b and round down''.

Example 1.3 (Floor)**.** $\lfloor 10/4 \rfloor = 2$ because $\overset{q}{\underset{\downarrow}{2}} \cdot \overset{b}{\underset{\downarrow}{4}} = 8 \leq \overset{a}{\underset{\downarrow}{10}} < 12 = (\overset{q}{\underset{\downarrow}{2}} + 1) \cdot \overset{b}{\underset{\downarrow}{4}}$. Intuitively, you may divide and slash off the fractional part: $10/4 = 2\not{5}$.

Definition 1.4 (Modulo)**.** Let $a, b \in \mathbb{N}$. If $a < b$, then $a \,\%\, b = a$. Otherwise, $a \,\%\, b$ is the result of repeatedly subtracting b from a until the resultant number is less than a.

Theorem 1.1 (Modulo and divisibility)**.** $b \mid a \iff a \,\%\, b = 0$ *(see Definition 1.1 for equivalence).*

Example 1.4 (Modulo)**.** $20 \,\%\, 7 = 6$ because $20 - 7 = 13$, $13 - 7 = 6 < 7$. Equivalently, it suffices to find a multiple of 7 that is less than 20 and subtract that: $7 \cdot 2 = 14 < 20$, and $20 - 14 = 6 < 7$.

[1] The flooring operator is a unary operator $\lfloor \cdot \rfloor$ (it accepts only one input, not two) and is fed x/y as input in this special case.

[2] Throughout this document, I use the convention that 0 is a natural number. The natural numbers are then $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$.

2 Original Algorithm

To compute the day of the week (''DOW'') for an arbitrary date, we first determine the DOW for the ''doomsday'' of a given month and year, from which we can manually count to the target day. The algorithm is as follows.

1. For the given century, determine the rotating century code c using the table below.

	⋮	⋮	⋮	⋮
	1400	1500	1600	1700
	1800	1900	**2000**	2100
	2200	2300	2400	2500
	⋮	⋮	⋮	⋮
c	**5**	**3**	**2**	**0**

2. Take the last two digits of the year, call it y. Find the largest multiple of 12 that is less than or equal to y from the multiples table, and call it $12m$.

m	0	1	2	3	4	5	6	7	8
$12m$	0	12	24	36	48	60	72	84	96

3. Compute the remainder $r = y - 12m$, and the floor $\lfloor r/4 \rfloor$ (Definition 1.3) from the table.

r	0	1	2	3	4	5	6	7	8	9	10	11
$\lfloor r/4 \rfloor$	0	0	0	0	1	1	1	1	2	2	2	2

4. Now we compute the sum (see Definition 1.4 for $(\cdot)\ \%\ 7$)

$$D = (c + m + r + \lfloor r/4 \rfloor)\ \%\ 7. \tag{2.1}$$

   D determines the day of the week of every ''doomsday'' for that year via the following table.

DOW	Sun	Mon	Tue	Wed	Thu	Fri	Sat
D	0	1	2	3	4	5	6

5. Locate the desired month's doomsday from the following table.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Month #	1	2	3	4	5	6	7	8	9	10	11	12
Day of month	3 | 4	28 | 29	14	4	9	6	11	8	5	10	7	12

   January and February are special cases: choose the second number if it is a leap year, and the first number otherwise. A given year Y is a leap year if it is divisible by 4 but not by 100, unless it is also divisible by 400 (see Definition 1.2 for divisibility). For example, the years 1704, 1708, 1712, etc. are leap years because they are divisible by four; but the year 1700 is not a leap year because it is divisible by 100 but not 400; finally, 1600 is a leap year because it is divisible by 400. Note that divisibility by four depends only on the final two digits and not the century, while divisibility by 100 and 400 depend only on the century.

6. Having determined the DOW of the doomsday for the month of the desired date, you can now count to a particular day of interest.

A recommended strategy for learning the algorithm is as follows.

1. Read and follow the example below.

2. Try following either the above abstract definition of the algorithm or the example below, but with a different date of your choosing, in your head.
3. Repeat this process but attempt to first actively recall each step and table before verifying your recollection in the document.
4. Once you have the steps and tables memorized, practice the algorithm to get faster.

3 Original Algorithm Example

Congress officially adopted the United States Declaration of Independence on July 4th, 1776, announcing that the 13 American colonies were free from British rule. *What was the day of the week of this event?*

1. The century code from the table is $c = 0$:

	⋮	⋮	⋮	⋮
	1400	1500	1600	**1700**
	1800	1900	2000	2100
	2200	2300	2400	2500
	⋮	⋮	⋮	⋮
c	**5**	**3**	**2**	$\rightarrow$ **0**

2. The last two digits of 17**76** are $y = \mathbf{76}$. From the multiples table,

m	1	2	3	4	5	**6**	7	8
$12m$	12	24	36	48	60	**72**	84	96

   we choose $m = 6$ and $12m = 72$ because $6 \cdot 12 = 72 \leq 76 < 84 = 7 \cdot 12$.

3. We compute the remainder and floor,

$$r = y - 12m = 76 - 72 = 4, \quad \lfloor r/4 \rfloor = 1,$$

   with the aid of the flooring table,

r	0	1	2	3	**4**	5	6	7	8	9	10	11
$\lfloor r/4 \rfloor$	0	0	0	0	**1**	1	1	1	2	2	2	2

4. Having collected $(c, m, r, \lfloor r/4 \rfloor)$, we now compute Equation (2.1).

$$D = (c + m + r + \lfloor r/4 \rfloor) \,\%\, 7 = (6 + 0 + 4 + 1) \,\%\, 7 = 11 \,\%\, 7 = 4.$$

   Hence, from the table, we conclude that the doomsdays of 1776 were Thursdays:

Weekday	Sun	Mon	Tue	Wed	**Thu**	Fri	Sat
D	0	1	2	3	**4**	5	6

5. According to the table, July's annual doomsday is July 11th:

Month	Jan	Feb	Mar	Apr	May	Jun	**Jul**	Aug	Sep	Oct	Nov	Dec
Month #	1	2	3	4	5	6	**7**	8	9	10	11	12
Day of month	3 | 4	28 | 29	14	4	9	6	**11**	8	5	10	7	12

6. From the algorithm, we conclude that July 11th, 1776 was a Thursday. This was exactly one week from July 4th, 1776 ($11 - 4 = 7$), so we conclude that Congress officially adopted the United States Declaration of Independence on a **Thursday**.

4 Modified Algorithm

I now present a modified algorithm which I find to be simpler and quicker. In essence, it is an alternative but equivalent method for computing the function $D(c, y)$ % 7.

1. Take the last two digits of the year, call it y. Determine the greatest multiple of 4 that is less than or equal to y, and call it q (the maximum $q = 4n$ for $q \leq y$ and $n \in \mathbb{N}$). q is the most recent leap year at that date. Formally, q is given by

$$y - 4 < q = 4\lfloor y/4 \rfloor \leq y \leq 99, \tag{4.1}$$

   but q can be immediately determined much more easily with the following method.

   To find q quickly, note that if the first digit of y is odd, then y is a multiple of 4 if and only if the second digit is 2 or 6; if the first digit of y is even, then y is a multiple of 4 if and only if the second digit is a multiple of 4.

2. Compute the remainder $r = y - q$ with $0 \leq r \leq 3$.

3. For the given century, determine the rotating century code c using the table below,

	⋮	⋮	⋮	⋮
	1400	1500	1600	1700
	1800	1900	**2000**	2100
	2200	2300	2400	2500
	⋮	⋮	⋮	⋮
c	**5**	**3**	**2**	**0**

   and compute $D = q/2 - (c + r)$ with $-8 \leq D \leq 48$.[3]

4. Choose the smallest multiple of 7 from the table that is greater than or equal to D, and call it $7n \geq D$:[4]

n	−1	0	1	2	3	4	5	6	7
$7n$	−7	0	7	14	21	28	35	42	49

   Then the day of the week for every doomsday for that year is given by the following table:

DOW	Sun	Mon	Tue	Wed	Thu	Fri	Sat
$7n - D$	0	1	2	3	4	5	6

   You do not need to memorize or take note of what n is.

5. Locate the desired month's doomsday from the following table as before.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Month #	1	2	3	4	5	6	7	8	9	10	11	12
Day of month	3 | 4	28 | 29	14	4	9	6	11	8	5	10	7	12

   For January and February, remember to select the second date if it is a leap year and the first if it is not a leap year.

6. Having determined the DOW of the doomsday for the month of the desired date, you can now count to a particular day of interest.

[3] Note the following fast method for computing $q/2$. Let $q = 10a + b$ represent q with digits a and b. If a is even, then so is b, and each digit of q can be separately divided by 2. If a is odd, then the first digit of $q/2$ is $(a - 1)/2$, and the second digit is $b/2 + 5$.

[4] Note also that, in the above table, there are always one positive instance of $7n$ with a particular odd first digit and two positive instances of $7n$ with a particular even first digit. For example, there are two multiples of 7 with first digit 2 (21 and 28) and one multiple of 7 with first digit 3 (35).

5 Modified Algorithm Example

Once again, consider the date July 4th, 1776. We wish to find the day of the week of this date. We follow the method as follows

1. $y = 76$, and 76 is a leap year because 7 is odd and the last digit is a 2 or a 6. Hence, $q = 76$.
2. $r = 0$ because $y = q \iff r = y - q = 0$.
3. The century code from the table is $c = 0$:

	⋮	⋮	⋮	⋮
	1400	1500	1600	**1700**
	1800	1900	2000	2100
	2200	2300	2400	2500
	⋮	⋮	⋮	⋮
c	**5**	**3**	**2**	$\rightarrow$ **0**

   $q = 76$ has an odd first digit, so the first digit of $q/2$ is $(7-1)/2 = 3$ and the second digit is $6/2 + 5 = 8$. Having determined that $q/2 = 38$, we compute that $D = q/2 - (c + r) = 38 - (0 + 0) = 38$.

4. The smallest multiple of 7 from the table that is greater than or equal to 38 is $7n = 42$:

n	-1	0	1	2	3	4	5	**6**	7
$7n$	-7	0	7	14	21	28	35	**42**	49

   We compute that $7n - D = 42 - 38 = 4$, so each Doomsday of 1778 is a Thursday according to the table:

DOW	Sun	Mon	Tue	Wed	**Thu**	Fri	Sat
$7n - D$	0	1	2	3	**4**	5	6

5. According to the table, July's annual doomsday is July 11th:

Month	Jan	Feb	Mar	Apr	May	Jun	**Jul**	Aug	Sep	Oct	Nov	Dec
Month #	1	2	3	4	5	6	**7**	8	9	10	11	12
Day of month	3 | 4	28 | 29	14	4	9	6	**11**	8	5	10	7	12

6. From the algorithm, we conclude that July 11th, 1776 was a Thursday. This was exactly one week from July 4th, 1776 ($11 - 4 = 7$), so we reaffirm, using the new method, that Congress officially adopted the United States Declaration of Independence on a **Thursday**.